

Packet Tracer - Configure IPv6 ACLs

Addressing Table

Device	Interface	IPv6 Address/Prefix	Default Gateway
PC1	NIC	2001:db8:1:10::10/64	fe80::1
PC2	NIC	2001:db8:1:11::11/64	fe80::1
Server3	NIC	2001:db8:1:30::30/64	fe80::30

Objectives

Part 1: Configure, Apply, and Verify an IPv6 ACL

Part 2: Configure, Apply, and Verify a Second IPv6 ACL

Instructions

Part 1: Configure, Apply, and Verify an IPv6 ACL

Logs indicate that a computer on the 2001:db8:1:11::0/64 network is repeatedly refreshing a web page. This is causing a Denial-of-Service (DoS) attack against **Server3**. Until the client can be identified and cleaned, you must block HTTP and HTTPS access to that network with an access list.

Step 1: Configure an ACL that will block HTTP and HTTPS access.

Configure an ACL named **BLOCK_HTTP** on **R1** with the following statements.

- Block HTTP and HTTPS traffic from reaching **Server3**.

```
R1(config)# deny tcp any host 2001:db8:1:30::30 eq www
```

```
R1(config)# deny tcp any host 2001:db8:1:30::30 eq 443
```

- Allow all other IPv6 traffic to pass.

Step 2: Apply the ACL to the correct interface.

Apply the ACL on the interface closest to the source of the traffic to be blocked.

Step 3: Verify the ACL implementation.

Verify that the ACL is operating as intended by conducting the following tests:

- = Open the **web browser** of **PC1** to <http://2001:db8:1:30::30> or <https://2001:db8:1:30::30>. The website should appear.
- = Open the **web browser** of **PC2** to <http://2001:db8:1:30::30> or <https://2001:db8:1:30::30>. The website should be blocked.
- = Ping from **PC2** to 2001:db8:1:30::30. The ping should be successful.

Part 2: Configure, Apply, and Verify a Second IPv6 ACL

The logs now indicate that your server is receiving pings from many different IPv6 addresses in a Distributed Denial of Service (DDoS) attack. You must filter ICMP ping requests to your server.

Step 1: Create an access list to block ICMP.

Configure an ACL named **BLOCK_ICMP** on **R3** with the following statements:

- Block all ICMP traffic from any hosts to any destination.
- Allow all other IPv6 traffic to pass.

Step 2: Apply the ACL to the correct interface.

In this case, ICMP traffic can come from any source. To ensure that ICMP traffic is blocked, regardless of its source or any changes that occur to the network topology, apply the ACL closest to the destination.

Step 3: Verify that the proper access list functions.

- Ping from **PC2** to 2001:db8:1:30::30. The ping should fail.
- Ping from **PC1** to 2001:db8:1:30::30. The ping should fail.
- Open the **web browser** of **PC1** to <http://2001:db8:1:30::30> or <https://2001:db8:1:30::30>. The website should display.

The screenshot displays the Cisco Packet Tracer interface. On the left, the 'Addressing Table' is visible, showing the following configuration:

Device	Interface	IPv6 Address/Prefix	Default Gateway
PC1	NIC	2001:db8:1:10::10/64	fe80::1
PC2	NIC	2001:db8:1:11::10/64	fe80::1
Server3	NIC	2001:db8:1:30::30/64	fe80::30

The main workspace shows a network topology with PC1, PC2, R1, R2, R3, and Server3. A window for PC1 is open, showing the 'Web Browser' application. The URL bar contains 'https://2001:db8:1:30::30'. The browser displays the 'Cisco Packet Tracer - Server3' website, which includes a welcome message and quick links to 'A small page', 'Captures', 'Image page', and 'Image'.